

PROJECT

5.9

VPC ENDPOINTS

NETWORKING SERIES

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Objective

Build a secure connection between your VPC and S3 without touching the open internet using a VPC endpoint. This keeps your data traffic private, fast, and cost effective inside AWS land.

Add bucket policies like secret rules to your S3 vault to control who can enter and what they can do. This ensures your treasures are safe while still being reachable by trusted cloud citizens.

Services

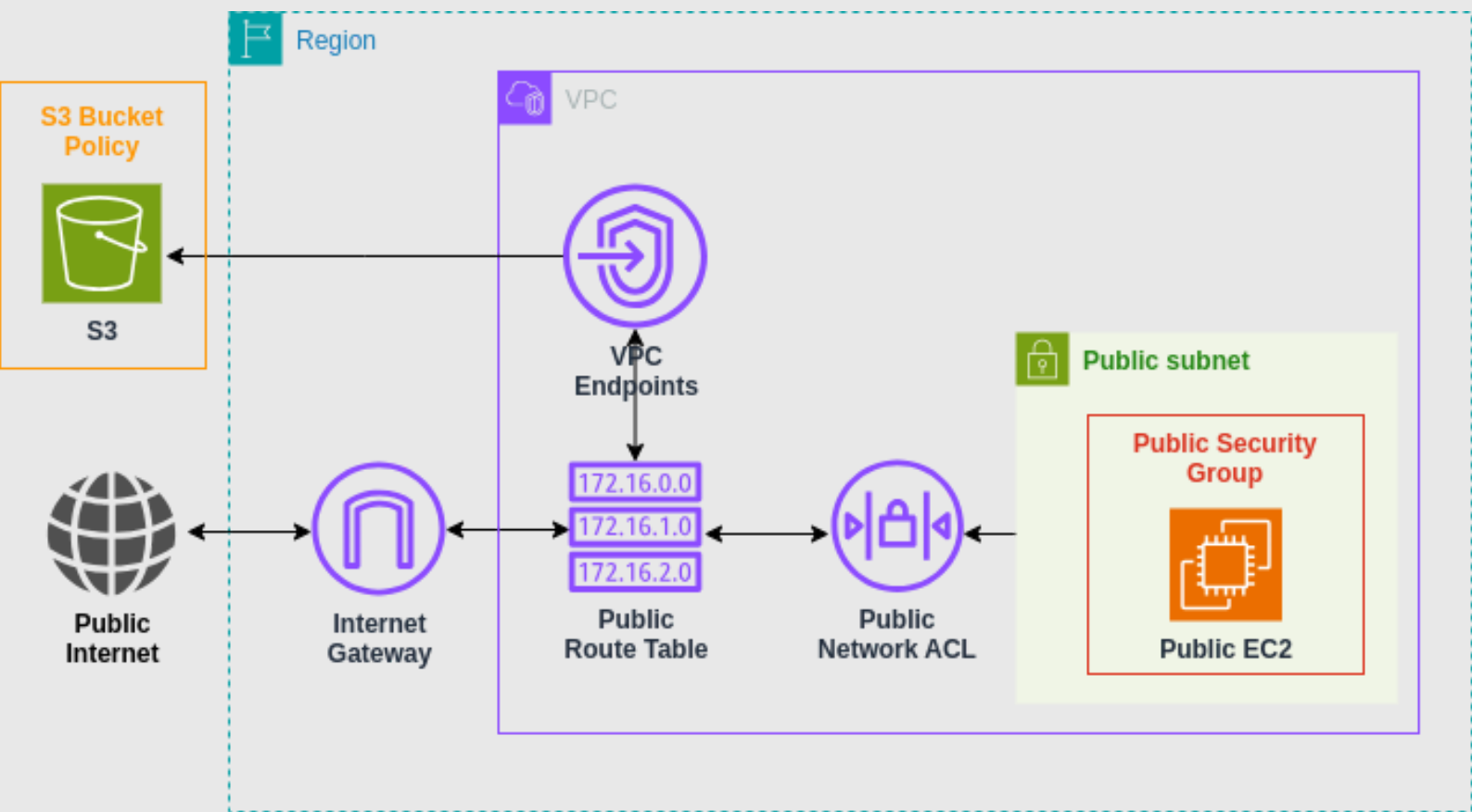
Amazon VPC - Creates a private network space to host your resources securely.

Amazon S3 - Stores your data like a cloud based vault or storage locker.

VPC Endpoint (Gateway) - Connects your VPC directly to S3 without using the internet.

S3 Bucket Policies - Controls access to your bucket with precise, rule-based permissions

Architecture Diagram



The VPC connects privately to S3 using a gateway endpoint, avoiding internet traffic. An S3 bucket policy ensures only this VPC can access the bucket. This setup boosts security and keeps data flow internal.

Steps to Implement

1. Create a VPC and S3 Bucket:

Start by launching your own VPC (your private cloud city) and create an S3 bucket (your treasure chest). Choose CIDR blocks like 10.0.0.0/16 for your VPC. This sets up your secure cloud space and a storage vault to protect.

2. Add a VPC Gateway Endpoint for S3:

Now, build a private road from your cloud city to your treasure chest by adding a Gateway VPC Endpoint. Select the S3 service and attach it to the route table of your subnet. This enables private, fast, and secure access without ever touching the internet.

3. Attach a Bucket Policy to S3:

Next, guard your treasure chest by writing a bucket policy that only allows access from your specific VPC. Mention the VPC endpoint ID in the policy for fine-grained control. Now, only your private road can reach the bucket, no outsiders allowed!

4. Test the Secure Connection:

Spin up an EC2 instance inside your VPC and try accessing the S3 bucket using the AWS CLI. Make sure the access is successful only through the VPC endpoint. If access from public internet fails congrats, your vault is sealed tight!

Story Time

VPC (Your Cloud City):

Imagine crafting your own cloud kingdom, wrapped with secure walls that's your VPC! Inside this private city, you place your EC2 instances, all safe and sound. These cloud citizens don't talk to the public internet unless you let them.

S3 Bucket (The Treasure Vault):

This is your precious vault where all digital scrolls and artifacts (data) are stored. It lives outside the castle but can be reached if allowed. Normally, you'd use open roads (internet) to get there, but we want a private tunnel instead!

VPC Endpoint (The Secret Tunnel):

Instead of using public highways, you create a magical tunnel from your city straight to the treasure vault. This is the VPC endpoint, secure, invisible, and exclusive! Now only your cloud city residents can access the S3 treasures directly.

Bucket Policy (The Guard's Rulebook):

To make the treasure vault extra secure, you give it a rulebook (bucket policy). It says: "Only the tunnel from our cloud city is allowed!" Anyone without the secret path is turned away, even if they know where the vault is.

Testing (Your Knight's Quest):

Now send a knight (EC2 instance) on a mission to retrieve scrolls from the vault. If he succeeds and intruders fail, your setup works! The kingdom is safe, the vault is private, and your network security is legendary!

BY
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THANK YOU!

